

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

III Semester of BSc Biological Sciences Examination March 2019

Course code & Name: BS309 Molecular Biology

15.03.2019

Friday

01:30 PM To 02:00 PM

Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I Choose the correct answer for the following questions.

20

1. Molecular biology involves study of
(i) DNA and RNA only (ii) DNA, RNA and Protein
(iii) Nucleic acids, Proteins and Lipids (iv) None of the above
2. RNA as genetic material is generally seen in
(i) Viruses (ii) Bacteria
(iii) Protozoa (iv) Mice
3. Double stranded model of DNA was proposed by
(i) Hershey and Chase (ii) Watson and Crick
(iii) Gregor Mendel (iv) Venkatraman Ramakrishnan
4. Which of the following RNA carries amino acid
(i) miRNA (ii) tRNA
(iii) mRNA (iv) None of the above
5. The non-variable 3' terminal sequence found in tRNA is
(i) CCG (ii) CCU
(iii) CCC (iv) CCA

6. DNA replication is
 - (i) Copying of DNA
 - (ii) Fragmentation of DNA
 - (iii) Secretion of DNA
 - (iv) Amplification of DNA
7. "Excision, recombination, mismatch" are associated with
 - (i) DNA replication
 - (ii) DNA repair
 - (iii) DNA duplication
 - (iv) DNA damage
8. DNA helicase
 - (i) Unwind DNA
 - (ii) Supercoils DNA
 - (iii) Fragments DNA
 - (iv) Repairs DNA
9. C-value refers to
 - (i) Amount of DNA
 - (ii) Replication of DNA
 - (iii) Structure of DNA
 - (iv) Isolation of DNA
10. Telomeres are found in
 - (i) DNA
 - (ii) RNA
 - (iii) Proteins
 - (iv) Lipids
11. Rolling circle mechanism of replication is found in
 - (i) Bacteria
 - (ii) Plants
 - (iii) Mice
 - (iv) Humans
12. Translation in molecular biology refers to synthesis of
 - (i) DNA
 - (ii) RNA
 - (iii) Proteins
 - (iv) Carbohydrates
13. Wobble hypothesis associated with
 - (i) Gene sequencing
 - (ii) Gene mutation
 - (iii) Genome Size
 - (iv) Genetic code
14. Select the right combination
 - (i) χ mechanism/ plasmid replication
 - (ii) α mechanism/ plasmid replication
 - (iii) β mechanism/ plasmid replication
 - (iv) θ mechanism/ plasmid replication
15. Total RNA present in the cell is known as
 - (i) RNAome
 - (ii) Genome
 - (iii) Transcriptome
 - (iv) Ribosome
16. Chaperone leads to protein folding True/ False
17. *trp* operon is an inducible system in bacteria True/ False
18. Linking number is a sum of twists and writhes True/ False
19. A nick is a single-strand break in a double-stranded nucleic acid True/ False
20. Plasmids are found in nucleus of yeasts True/ False

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III Semester of BSc Biological Sciences Examination March 2019

Course code & Name: BS309 Molecular Biology

15.03.2019

Friday

02:01 PM To 04:30 PM

Maximum Marks: 50

Instructions:

1. Section I and II must be attempted in TWO ANSWER SHEETS
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. how necessary calculations.

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SECTION -I

Q - II Answer the following questions as directed **20**

1. Discuss **ANY ONE** **5**
 - (i) Hershey–Chase experiment
 - (ii) Griffith's experiment
2. Describe post translational modification of proteins in eukaryotes **5**

OR

Describe post transcriptional modification in eukaryotes
3. Attempt **ANY TWO** **10**
 - (i) Genome organization in eukaryotes
 - (ii) Describe the process of transcription in prokaryotes
 - (iii) Protein synthesis in prokaryotes

SECTION – II

Q-III Answer the following questions as directed **30**

1. Describe DNA replication either in **5**
 - (i) *Prokaryotes*

OR

 - (ii) *Eukaryotes*
2. Describe **ANY FIVE** **25**
 - (i) miRNA
 - (ii) DNA Repair
 - (iii) *trp* operon
 - (iv) Molecular switch in phage lambda
 - (v) *lac* operon
 - (vi) Gene Mutation
 - (vii) Chromatin and forms of chromatin